

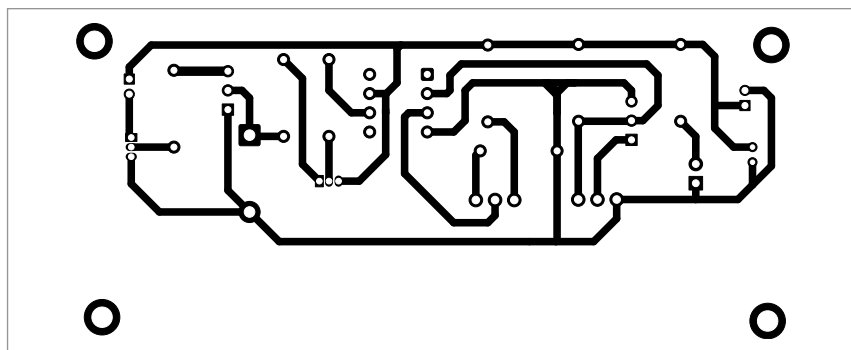


Fig. 3: Actual-size PCB layout

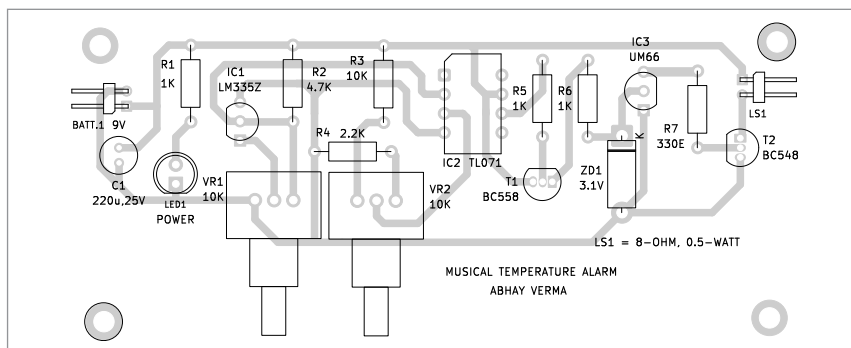


Fig. 4: Component side of the PCB

In this condition, the PNP transistor T1 is off, and no supply voltage is applied to the UM66 melody IC. Consequently, the melody generator remains inactive. Transistor T2 also remains non-conducting, and the loudspeaker produces no sound.

At high temperature (above the set threshold). As the temperature rises above the preset threshold, the LM335 output voltage exceeds the reference voltage. The TL071 comparator output then goes low. In this low-output condition, PNP transistor T1 conducts.

Once T1 conducts, it supplies power to the UM66 IC through the Zener-regulated supply. The UM66 immediately starts generating a preprogrammed musical signal at its output. This low-level audio signal is amplified by transistor T2 (BC548) and delivered to the loudspeaker, producing an audible melody as a temperature alarm.

Construction and testing

Assemble the components on a general-purpose PCB or design a small

custom PCB layout to keep the circuit compact. Place the LM335 temperature sensor away from heat-generating components, such as transistors and the voltage regulator, to ensure accurate temperature measurements.

Mount the TL071 and UM66 in IC sockets to allow easy replacement or testing. The LED should be positioned at the edge of the board for visibility. The VR1 potentiometer should be easily accessible for

adjusting the triggering temperature.

Ensure the Zener diode and UM66 are connected correctly; incorrect polarity can damage these components. A small 6-ohm, 0.5-watt speaker can be used for sound output, or a piezo buzzer may be connected if preferred.

For the enclosure, a small plastic box with ventilation holes near the sensor may be used. If the circuit is intended for outdoor or high-humidity environments, the PCB should be coated with a conformal coating or housed in a ventilated but protected casing.

Testing. To test the circuit, first power it on using a 9V battery. The LED should glow, indicating that the power supply is active. Begin the initial calibration by adjusting preset VR1 until the alarm remains silent at normal room temperature (approximately 25°C).

Next, gently heat the LM335 temperature sensor using a hair dryer or by bringing it close to a warm object. As the temperature rises, observe the speaker's output. When the temperature crosses the preset threshold, the musical melody should begin playing.

If the alarm activates too early or too late, fine adjustments can be made by slightly rotating VR1 until the desired temperature response is achieved. VR2 may also be adjusted if the sensor response appears offset or inaccurate.

Allow the sensor to cool naturally after testing. The music should stop automatically once the temperature falls below the set threshold, confirming correct circuit operation. For improved stability, a small heat sink may be attached to the sensor to ensure smoother thermal response and more consistent triggering. **EFY**

Bonus. You can watch the video of the tutorial of this DIY project at <https://www.electronicsforu.com/videos-slideshows/make-your-own-musical-temperature-alarm>

PARTS LIST

Semiconductors:

IC1	- LM335Z temperature sensor
IC2	- TL071 op-amp
IC3	- UM66 melody generator IC
T1	- BC558 PNP transistor
T2	- BC548 NPN transistor
ZD1	- 3.1V zener diode
LED1	- 5mm LED

Resistors (all 1/4-watt, ±5% carbon):

R1, R5, R6	- 1-kilo-ohm
R2	- 4.7-kilo-ohm
R3	- 10-kilo-ohm
R4	- 2.2-kilo-ohm
R7	- 330-ohm
VR1, VR2	- 10-kilo-ohm potmeter

Capacitors:

C1	- 220µF, 25V electrolytic
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Miscellaneous:

BATT.1	- 9V battery
LS1	- 8-ohm, 0.5-watt speaker

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